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MILITARY SPECIFICATION

PUMPS, AUTOMOTIVE FUEL DISPENSING, (SERVICE STATION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric-motor-driven automotive fuel dispensing service station pumps.

1.2 Classification. Service station pumps shall be the following sizes as specified (see 6.2):

Size 1 - 12 to 15 gallons per minute (gpm) capacity.
Size 2 - 22 to 25 gpm capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motor, Alternating-Current, Fractional and Integral Horsepower (500 HP and Smaller).
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 4930

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-N-52110	- Nozzles, Fuel and Oil Servicing, Non-Automatic Shut-Off and Nozzles, Fuel Servicing, Automatic Shut-Off.
MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/7	- Screw-Thread Standards for Federal Services, Section 7, American Standard Pipe Threads (except Dryseal and Hose Coupling Types).
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/UL87 - Power Operated Dispensing Devices for Petroleum Products.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook 44 - Specifications, Tolerances, and Other Technical Requirements for Commercial Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J492 - Rivets and Riveting.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

SUPERINTENDENT OF DOCUMENTS

40 CFR Part 80 - Regulation of Fuel and Fuel Additives.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The pumps shall be self-contained, power-operated fuel dispensing service station pumps as specified herein and in Underwriters' Laboratories Standard for Safety ANSI/UL87.

3.2 First article (first produced pumps). The contractor shall furnish one or more pumps as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce pumps that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein and in ANSI/UL87. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Threads. All screw threads shall conform to FED-STD-H28.

3.5 Design and safety.

3.5.1 Fire and casualty hazards. Each pump shall bear the Listing Mark of the Underwriters' Laboratories, Inc., as evidence that the pump to which it is attached has been produced under the Listing and Follow-Up Services of the Laboratories and that the pump complies with the requirements for self-contained, power-operated dispensing divices as set forth in ANSI/UL87. In lieu of the Underwriters' Listing Mark, the contractor may submit proof, which shall be subject to approval of the contracting officer, that each pump has been approved for the purpose intended by an independent nationally recognized testing agency indicating conformance to the standards and requirements established by this specification. Agency approval shall not absolve the contractor from compliance with the requirements of this specification.

3.5.2 Federal regulations. Prior to delivery of the first produced pumps, the contractor shall certify that the pumps comply with all applicable requirements of 40 CFR Part 80.

3.5.3 National Bureau of Standards. The pumps shall comply with all design and operational requirements of NBS Handbook 44 for retail liquid fuel dispensing devices.

3.6 Ease of maintenance. All major assemblies and parts subject to wear shall be accessible for adjustment, repair, and replacement without the removal of other major assemblies. Common hardware and fastening devices such as bolts, nuts, screws, and similar items shall be held to a minimum number

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of types and sizes. All fasteners shall be corrosion-resistant material or shall be treated to be corrosion-resistant. All maintenance, assembly, or disassembly operations shall be accomplished with common tools and those special tools which are furnished with the pump unit.

3.7 Performance. The pumps shall perform in accordance with 3.19 when dispensing gasoline conforming to MIL-G-3056, operating at standard temperature with a suction lift of 6.5 to 7.5 feet (measured from fluid level to center line of pumping unit). The pumps shall not allow any leakage or seepage of fuel between the inlet and outlet of the pumps.

3.8 Pumping units. The pumping units shall be high vacuum, rotary type. The pumping unit may be connected to the electric motor by either a V-belt or a cog-timing belt, or may be directly coupled with the electric motor. Unless otherwise specified (see 6.2), the pump shall provide for emergency manual operation at a minimum delivery rate of 3.0 gallons per minute (gpm).

3.9 Strainer. The pumping units shall be equipped with a suction-side strainer having a 50 by 60 or finer mesh, reinforced, corrosion-resistant screen for use with gasoline. It shall have an effective flow area at least three times the minimum suction inlet pipe area. It shall be easily removable for cleaning or replacement.

3.10 Air separator and release. The pumping units shall be equipped with an air eliminator unit in accordance with NBS Handbook 44. Venting shall comply with ANSI/UL87.

3.11 Check valve. The pumps shall be provided with a check valve to keep the discharge lines full of fuel.

3.12 Pressure relief devices.

3.12.1 Bypass valve. Each pumping unit shall be equipped with a manual or self-adjusting bypass valve capable of bypassing the entire output of the pump.

3.12.2 Fuel expansion pressure relief. Each pump shall be equipped with a device for automatically relieving pressure built up within the system caused by expansion of the fuel. The device shall be designed to prevent or relieve pressures and thermal expansion pressures in the system in excess of 50 pounds per square inch (psi).

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3.13 Measuring devices.

3.13.1 Flow meter. The measuring unit for the pumps shall be a volumetric, positive-displacement flow meter. The meter shall be capable of withstanding a maximum pressure of not less than 50 psi. Adjustments, performance requirements, and acceptance tolerances shall comply with NBS Handbook 44 for retail liquid fuel dispensing devices.

3.13.2 Register, noncomputing. The pumps shall be equipped with a noncomputing register having a minimum of four rotor set-back numeral wheels visible from the front side of the dispenser. The register shall show readings in fractions of a gallon up to at least 999.9 gallons. The numerals shall be not less than 0.344 inch high and shall be of a contrasting color with the rotor setback numeral wheels background color. Unless otherwise specified (see 6.2), the register shall show US gallons delivered, and shall be in agreement with the flow meter.

3.13.3 Totalizer. The pumps shall be provided with a nonset-back totalizer capable of recording up to not less than 100,000 gallons delivered, in tenths of a gallon, i.e. 99,999.9.

3.13.4 Set-back mechanism. The pumps shall be provided with a manual set-back mechanism located in a position convenient to the pump operator. The mechanism shall be designed so that the numeral wheels of the register must be returned to zero reading before a new delivery can be made. Operation and design of the mechanism shall be in accordance with NBS Handbook 44.

3.14 Electrical equipment.

3.14.1 Motors. Unless otherwise specified (see 6.2), the electric motors shall be single phase, continuous or intermittent duty, operable from either 120 or 240 volt alternating current (ac), 60 Hertz (Hz) source at a synchronous speed of 1700 to 1800 revolutions per minute (rpm). Source of voltage selection shall be accomplished by either a built-in switch (see 3.14.1.1) or by repositioning the external source wires (see 3.14.1.2), at the option of the contractor. The motors shall conform to CC-M-1807, Design L (explosion-proof), except that, the temperature rise of a motor shall not exceed 55° C during sustained operation for 30 minutes, plus or minus one minute. An automatic thermal overload mechanism shall be provided with automatic reset.

3.14.1.1 Source voltage selection switch. A source voltage selection switch, when provided, shall allow quick selection of either 120 or 240 source voltage, with the switch position for each voltage clearly marked. A removable screw shall maintain the selected position and shall prevent accidental repositioning.

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3.14.1.2 Source wiring. Motors not equipped with a built-in source voltage switch shall be equipped with instruction plates or diagrams, including warnings or cautions, clearly indicating the method of connection for each source voltage. The plates shall be attached by screws, bolts, or rivets in a conspicuous, protected location.

3.14.2 Control switches. All necessary switches for controlling the electrical equipment shall be provided and shall be in accordance with ANSI/UL87.

3.14.3 Wiring. The pumps shall be completely wired and ready for connection to a power circuit.

3.15 Hose, couplings and nozzles.

3.15.1 Hose and couplings. Hose and couplings shall be the type normally furnished with the pumps and shall comply with the requirements of ANSI/UL87 and NBS Handbook 44. Couplings shall have NPT threads conforming to FED-STD-H28/7.

3.15.2 Nozzle. Unless otherwise specified (see 6.2), the nozzle shall conform to MIL-N-52110, Type II, Size 1.

3.15.2.1 Arctic-type nozzle. When specified (see 6.2), an arctic-type nozzle conforming to MIL-N-52110, Type I, Size 1, shall be provided with the pump.

3.15.2.2 Nozzles for use with gasoline. When pumps are to be used to dispense gasoline, nozzles shall be for use with leaded or unleaded gasoline as specified (see 6.2), with curved or straight spout as specified (see 6.2), in accordance with 40 CFR Part 80, and as specified herein.

3.15.2.2.1 Leaded gasoline nozzles. The terminal end of nozzle spouts for use with leaded gasoline shall have an outside diameter not less than 0.930 inches.

3.15.2.2.2 Unleaded gasoline nozzles. The terminal end of nozzle spouts for use with unleaded gasoline shall have an outside diameter not more than 0.840 inches. The delivery end of the spout shall include a straight section not less than 2.50 inches long. The retaining spring shall terminate 3.0 inches from the terminal end of the spout.

3.16 Base, pedestal, and housing. The base, pedestal, and housing shall be in accordance with ANSI/UL87 and as specified herein.

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3.16.1 Base and pedestal. The complete pumping unit shall be mounted on a steel base and pedestal of welded or bolted construction. Supporting brackets shall be iron or steel, welded or bolted to the frame.

3.16.2 Housing. The pump connections shall be enclosed in a steel housing designed to minimize entry of dust, moisture, water, and shall afford accessibility to all interior components. Each panel or door of the housing shall be capable of being locked or secured in position. The housing shall provide for receiving the hose nozzle tip into the nozzle locking and support device.

3.17 Identification marking. The pumps shall be identified in accordance with ANSI/UL87, and MIL-STD-130.

3.18 Treatment and painting. The pumps, including all components, shall be cleaned and finished in accordance with the contractor's standard practice. Unless otherwise specified (see 6.2), the exterior finish of the pump housing except for inherently corrosion-resistant surfaces, shall be gloss yellow conforming to FED. STD. No. 595, Color 13538, with only visual comparison being required.

3.18.1 Register dial. The color of the dial shall be at the option of the contractor.

3.19 Size 1 and Size 2 pump requirements. Size 1 and Size 2 pump performance, number of pumping units, pump connections, motor horsepower, flow meter, and hose dimensions, shall be in accordance with Table I.

TABLE I. Pump size requirements.

Requirement		Size 1	Size 2
Performance (gpm delivery)		12-15	22-25
No. of pumping units		1	1 or 2
Pump connections (inches internal NPT)	Inlet	1-1/2	1-1/2
	Outlet	3/4	1
Motor horsepower (minimum rating)		1/3	3/4
Flow meter	Size (inches)	3/4	1
	Rated capacity (gpm)	15	25
Hose	Length (feet)	10	12
	Nominal inside dia. (inches)	3/4	1
	Couplings (inches NPT)	3/4	1

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3.20 Workmanship.

3.20.1 Castings and forgings. All parts, components, and assemblies of the pumps which include castings and forgings shall be clean from harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.20.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.20.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the pumps to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pumps shall be at the option of the contractor.

3.20.4 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.20.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.20.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.20.7 Painting. The paint shall not be blistered, peeled, chipped, scratched, or contain imbedded foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced pump inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced pump inspection.

4.3.1 Examination. The first produced pumps shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced pumps shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Level II for examination and Level S-2 for tests.

4.4.2 Examination.

4.4.2.1 Individual. Each pump shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 6.5 percent defective for major defects and 10 percent defective for minor defects.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pumps shall be examined as specified herein for the following defects:

Critical

1. UL approved or equivalent certification not provided.

Major

101. Incorrect size pump.
102. Certification of compliance with 40 CFR Part 80 not provided.
103. Components missing or not properly assembled.
104. Materials not as specified.
105. Ease of maintenance provisions not as specified.
106. Pumping units of incorrect type, or not connected to motor as specified.
107. Emergency manual operation not provided for, when required.
108. Strainer not as specified.
109. Source voltage selection switch or source wiring instructions, not as specified.
110. Hose and nozzle not as specified.
111. Base, pedestal, and housing not as specified.
112. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Before testing, the pump shall be connected to a container of gasoline conforming to MIL-G-3056, Type I or II, or to a similar solvent whose comparative properties are known. Connection shall be in accordance with 3.7. Connections in the fuel flow path shall be checked for tightness. Run-in the pump as follows: (1) Run pump at full discharge to dispense from 90 to 125 gallons of test medium so as to fully prime pump and eliminate all air. (2) Near completion of run-in period, check discharge capacity of pump for compliance with 3.7 and 3.19. (3) At completion of the run-in period, close nozzle and reset counter to zero.

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4.5.2.2 Operation and design. Operate the pump to determine conformance to ANSI/UL87 and NBS Handbook 44. Nonconformance to the operational design requirements of ANSI/UL87 for self-contained, power-operated, fuel dispensing service station pumps, or nonconformance to the operational or design requirements of NBS Handbook 44 for commercial weighing and measuring devices shall constitute failure of this test.

4.5.2.3 Performance. The pump shall be run-in as specified in 4.5.2.1 and its discharge capacity checked for compliance with 3.7 and 3.19. The flow meter shall be adjusted to within the acceptance tolerance limits set by NBS Handbook 44. Nonconformance to 3.7 or 3.19 shall constitute failure of this test.

4.5.2.4 Emergency manual operation. When emergency manual operation is provided for, test as follows: (1) Set pump up for manual operation. (2) Manually operate the pump discharging gas for 30 seconds. A delivery rate of less than 3 gpm shall constitute failure of this test.

4.5.2.5 Bypass valve efficiency. Test as follows: Operate the pump for 30 seconds, randomly varying the nozzle control setting between full and zero discharge. Any audible straining or lugging of the pump motor shall constitute failure of this test.

4.5.2.6 Pump performance, set-back mechanism, register, and totalizer. Test as follows: (1) Operate the pump at full discharge for one timed minute. (2) Record and compare the readings of both the noncomputing register and the totalizer. (3) Turn pump off and insert nozzle in pump nozzle locking device. (4) Without operating register set-back, remove nozzle and attempt to deliver gasoline. Noncomputing register readings not within specified gpm delivery range for size of pump, register readings not in agreement with totalizer increase, or ability to deliver gas without setting back register to zero shall constitute failure of this test.

4.5.2.7 Motor temperature rise. Test as follows: (1) Turn pump motor off and allow to stand idle for 15 minutes. (2) Record temperature of pump motor casing. (3) Turn pump motor on and allow pumping unit to circulate test fluid (on bypass) for a period of 30 minutes, plus or minus 1 minute. (4) Record temperature of pump motor casing. A temperature rise greater than 55° C shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

113.	Materials, processes, methods and containers not as specified for Level A or B.
114.	Disassembly not as specified for Level A.
115.	Drive belts not released from tension for Level A.
116.	Glass surfaces not protected as specified for Level A.
117.	Fiberboard boxes not closed as specified for Level A.
118.	Contents of crates not waterproofed as specified for Level A or B.
119.	Strapping not as specified for Level A or B.
120.	Blocking, bracing, and anchoring not in accordance with the referenced document for Level A or B.
121.	Marking missing, illegible, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Protruding and detachable components vulnerable to damage and that would otherwise increase cubage if not removed, shall be disassembled from the pump. Any removed bolts, nuts, screws, pins, and washers shall be reinstalled in mating parts and secured to prevent their loss. Parts of the pump that are free to move shall be secured in a fixed position to prevent movement while in transit.

5.1.1.2 Cleaning. Surfaces requiring the application of a preservative shall first be cleaned (Process C-1) and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Preservatives. Preservatives shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.4 Unprotected metal surfaces. All exposed unprotected metal surfaces requiring the application of a preservative shall be coated with Type P-1 preservative. Surfaces requiring a preservative where removal of the preservative is required prior to operation of the pump shall be coated with Type P-6 or P-11 preservative. Exterior surfaces coated with Type P-6 or P-11 preservative shall be wrapped or covered with barrier materials conforming to MIL-B-121, Type I, Grade A, Class 2 and secured in place with tape conforming to PPP-T-60, Type IV.

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5.1.1.5 Drive belts and pulleys. Drive belts shall be released from tension. Unprotected pulley grooves shall be coated with a primer conforming to TT-P-664.

5.1.1.6 Glass surfaces. All glass surfaces shall be protected with a double thickness of fiberboard conforming to PPP-F-320, Class Weather-Resistant, Variety DW and secured in place with tape conforming to PPP-T-60, Type IV.

5.1.1.7 Openings. All openings into the pump shall be sealed with tape conforming to PPP-T-60, Type IV.

5.1.1.8 Consolidation. All removed components shall be consolidated in a fiberboard box conforming to PPP-B-636, Class Weather-Resistant. The contents shall be cushioned to prevent movement and damage. The box shall be closed (Method V) in accordance with the appendix to the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The dispensing pump and consolidated package shall be packed in a crate conforming to MIL-C-52950, Style A, Type V, or MIL-C-3774, Type I, Style A or in a box conforming to PPP-B-601, Overseas Type. The pump shall be packed in an upright position. The contents of crates shall be waterproofed with shrouds in accordance with the appendix to the applicable crate specification. Closure and strapping shall be in accordance with the appendix to the applicable container specification. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186.

5.2.2 Level B. The dispensing pump and consolidated package shall be packed as specified for Level A in 5.2.1 except boxes conforming to PPP-B-601 shall be Domestic Type.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Pumps covered by this specification are intended to be permanently connected to a supply source at service stations or other vehicle servicing facilities to refuel automobiles, trucks, and buses.

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6.1.1 Size 1 pump. The Size 1 pump is intended for use where the majority of vehicles to be refueled have gasoline tank capacities of less than 100 gallons.

6.1.2 Size 2 pump. The Size 2 pump is intended for use where the majority of vehicles to be refueled have gasoline tank capacities of more than 100 gallons. It is not intended for filling bulk fuel transporters or tank trucks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of pump required (see 1.2).
(c) The time frame required for submission of first produced pumps, and number of pumps required (see 3.2).
(d) When provision for emergency manual operation is not required (see 3.8).
(e) When the register shall show other than US gallons (see 3.13.2).
(f) Electric motor characteristics if other than as specified (see 3.14.1).
(g) Type of nozzle required (see 3.15.2 and 3.15.2.1).
(h) Whether pump nozzles shall be for use with leaded or unleaded gasoline (see 3.15.2.2).
(i) Whether pump nozzle spout shall be curved or straight (see 3.15.2.2).
(j) Color of exterior finish when other than as specified (see 3.18).
(k) Welding process required when other than as specified (see 3.20.3).
(l) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced pumps. Any changes or deviations of production pumps from the approved first produced pumps during production will be subject to the approval of the contracting officer. Approval of the first produced pumps will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of pumps.

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6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - YD

Review activity:
DLA - CS

User activities:
Army - CE
Navy - MC

Preparing activity:
Army - ME

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-10406G Pumps, Automotive Fuel Dispensing, (Service Station)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED *PROBLEMS OR* REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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